

BDNSI

Bangladesh National Security Information

PROJECT MASTER DOCUMENTATION

Documentation Version: 1.0

Audit Date: 2026-08-24

Audited By: Antigravity AI Architect

Based On: Direct source code inspection

Project Path: c:\BDNSI

Live URL: <https://nenobet.live>

Completion: ~70-75%

PROJECT MASTER DOCUMENTATION

BDNSI — Bangladesh National Skills Institute Management System

Documentation Version: 1.0 **Audit Date:** 2026-08-24 **Audited By:** Antigravity AI Architect **Based On:** Direct source code inspection of `c:\BDNSI` **Completion Estimate:** ~70–75%

TABLE OF CONTENTS

- 1. Executive Summary
- 2. Project Purpose
- 3. Current Status
- 4. Technology Stack
- 5. Complete Project Structure
- 6. System Architecture
- 7. Module Inventory
- 8. Panel Inventory
- 9. Page & Route Inventory
- 10. Database Architecture
- 11. API Documentation
- 12. Authentication
- 13. Authorization
- 14. Business Workflows
- 15. Frontend Architecture
- 16. Backend Architecture
- 17. State Management
- 18. File Management

19. Notifications & SMS
 20. Payment System
 21. Integrations
 22. Security Audit
 23. Performance Audit
 24. Testing Audit
 25. Technical Debt
 26. Bugs & Issues
 27. Missing Features
 28. Recommended Improvements
 29. Final Architecture
 30. Development Roadmap
 31. AI Agent Rules
 32. Final Audit Summary
-

1. Executive Summary

BDNSI is a production Laravel 8 + React (Inertia.js) **multi-tenant Institute Management System** deployed on Hostinger at `https://nenobet.live`. The system serves training institutes across Bangladesh, enabling student enrollment, academic result management, document generation, online payment, and multi-portal access for admins, staff agents, centre operators, and students.

Approximately **70-75% of the system is production-ready**. The core academic lifecycle (enroll → result → document) is solid. Gaps exist in the online examination module, SMS notification wiring, and battle-tested payment verification.

Two critical security vulnerabilities must be patched immediately: an unauthenticated `/live\_deploy` route and hardcoded SSH credentials in deployment scripts.

2. Project Purpose

What It Does

- Manages vocational training institute operations in Bangladesh

- Supports 3 course types: Regular (1-3 months), Short Course (4-24 months), Diploma (25-48 months)
- Issues digital certificates, ID cards, admit cards, marksheets
- Tracks multi-centre student enrollments with tenant isolation
- Manages a sales agent (staff) referral and KPI system
- Provides public result verification by roll/registration number
- Accepts online payments via bKash and SSLCommerz

Who Uses It

User Type	Panel	Purpose
Super Admin	Admin Panel	Full system control
Sub-Admin	Admin Panel	Role-limited management
Staff/Agents	Staff Portal	Manage own students, track KPIs
Centre Operators	Centre Dashboard	Manage centre's students
Students	Student Portal	View results, download documents
Public	Website	Browse courses, verify results, contact

3. Current Status

Area	Status	Notes
Public Website	85% Complete	All major sections working
Admin Panel	80% Complete	Most CRUD complete
Student Portal	65% Complete	Docs need testing
Staff Portal	60% Complete	Core flow working
Centre Portal	55% Complete	Data gaps
Payment System	70% Integrated	Not battle-tested
SMS Notifications	30%	Drivers exist, not wired
Online Examination	40%	Backend partial, UI incomplete
Document Builder	75% Complete	Template + bulk PDF working
Laravel Telescope	100%	Secured, deployed
Testing (PHPUnit)	31/31 Passing	All passing
CI/CD	60%	Deploy scripts work, no GitHub Actions

4. Technology Stack

Backend

Technology	Version	Role
PHP	^8.2	Server language
Laravel	^8.x	MVC framework
Inertia.js server	^0.6	SPA bridge
Sanctum	^2.11	API auth
Laratrust	^7.1	RBAC
Telescope	4.6	Debug monitoring
Intervention/Image	^2.7	Image processing
Simple-QRCode	^4.2	QR generation
GuzzleHTTP	^7.0	External API calls

Frontend

Technology	Version	Role
React	^19.2.7	UI framework
Inertia.js React	^0.8.1	SPA adapter
Vite	^8.2	Build + HMR
Tailwind CSS	^3.0	Styling
Recharts	^3.10	Charts
jQuery + DataTables	^3.6 / ^1.11	Admin tables
Lucide React	^1.33	Icons

Database

- MySQL (production/local)
- SQLite (testing only)
- Eloquent ORM

Infrastructure

- Hosting: Hostinger VPS (145.79.212.19)
 - Domain: nenobet.live
 - Local Dev: XAMPP (Windows)
 - CI/CD: SSH/SFTP scripts (Node.js)
 - Version Control: GitHub
-

5. Complete Project Structure

c:\BDNSI/

├─ app/

| ├─ Casts/ ImageField cast

| ├─ Console/ Scheduler (Kernel.php – telescope:prune daily)

| ├─ Enums/ 9 PHP enums (StudentStatus, Gender, CourseType, etc.)

| ├─ Events/ PaymentSucceeded

| ├─ Http/

| | ├─ Controllers/

| | | ├─ Admin/ 38 admin controllers

| | | ├─ Staff/ 5 controllers

| | | ├─ Student/ 3 controllers

| | | └─ Auth/ Centre auth

| | └─ Middleware/ 14 middleware files

| | └─ Requests/ Form validation classes

| └─ Jobs/ 3 queued jobs (BulkDoc, PaymentSMS, StudentSMS)

```
|   ├── Lib/           Image.php utility + helpers.php (autoloaded)

|   ├── Listeners/     UpdateStudentFinancialStatus

|   ├── Models/        35 Eloquent models

|   ├── Observers/     StudentObserver, ResultObserver

|   ├── Providers/     5 service providers + TelescopeServiceProvider

|   ├── Scopes/        CenterScope (tenant isolation)

|   ├── Services/      DocumentGenerator, PdfEngine, SMS drivers

|   └── Traits/        BelongsToStaff, ClearsFrontendCache, DeletesImage

└── config/            Laravel config files (auth.php, telescope.php, etc.)

└── database/

|   ├── migrations/    75 migration files (2011-2026)

|   ├── seeders/

|   └── factories/

└── resources/

|   ├── js/
```



```
| | └─ Components/ SmartScanner (Gemini OCR), Document builder

| | └─ Layouts/      5 layouts (Admin, Staff, Student, Centre, Frontend)

| | └─ Pages/        70+ React page components

| └─ views/          Blade templates (admit card, certificate, transcript)

└─ routes/

    └─ web.php        Public + centre user routes (145 lines)

    └─ admin.php      Admin routes (201 lines)

    └─ staff.php      Staff routes (50 lines)

    └─ student.php    Student routes (50 lines)

    └─ auth.php       Centre auth routes

└─ tests/

    └─ Feature/       9 PHPUnit test files (31 tests total)

    └─ e2e/           16 Playwright spec files

└─ PROJECT_DOCUMENTATION/ This documentation folder
```

6. System Architecture

Request Lifecycle

Browser Request

- Hostinger LiteSpeed → public/index.php
- Bootstrap → Http Kernel
- Web Middleware: throttle, cookies, session, referral capture, CSRF, bindings, language, Inertia handler
- Route Matching (web/admin/staff/student)
- Route Middleware: auth:guard, module:toggle, throttle
- Controller → Service → Eloquent Model → MySQL
- Inertia::render('PageComponent', \$props)
 - + HandleInertiaRequests::share() adds:
 - auth.user/admin/staff/student, flash, site_config,
 - footer_links, footer_logos, locale, app_url
- React renders page via Inertia SPA

Authentication (4 Guards)

web guard → User model → users table → /login

admin guard → Admin model → admins table → /admin/login

student guard → Student model → students table → /students/login

staff guard → Team model → teams table → /staff/login

All guards: session-based driver, bcrypt passwords

Tenant Isolation

```
CenterScope (global scope on Student + Result)
  When auth('web')->user() has center_id:
    → Automatically WHERE center_id = {user.center_id}
  Admin guard: bypasses automatically
  Staff guard: BelongsToStaff trait (filters by team_id)
```

Feature Flag System

```
SiteConfig table → 10 toggle columns (toggle_result_verify, etc.)

CheckModuleEnabled middleware → SiteConfig::isEnabled($flag)

If disabled → Inertia::render('Frontend/ComingSoon')
```

7. Module Inventory

COMPLETE MODULES

Student Enrollment | Centre Management | Course (Subject) Management |
Session Management | Result Management | Document Templates |
Document Generation | Grade Scales | License Management |
Notice Board | Slider/Banner | Team/Staff Management |
Team KPI/Performance | Site Config/CMS | Footer Management |
Translation | Sponsor Management | YouTube Videos |
WhatsApp Links | Contact Inquiries | Payment Gateway Config |
SMS Gateway Config | User Management | Sub-admin Management |
Referral Tracking | Laravel Telescope |

PARTIAL MODULES

- Online Examination (backend models done, student exam UI incomplete)
- Payment System (integrated, not production-tested)
- Financial Tracking (manual records, not linked to online transactions)
- Registration Review (endpoint exists, workflow unclear)
- Diploma Management (minimal CRUD only)
- Sub-admin RBAC (role assignment done, enforcement gaps)
- Database Backup (controller exists, no schedule)
- Bulk PDF (job exists, queue worker not set up on production)
- Student Documents (routes exist, template linking unverified)
- SMS Notifications (drivers + jobs exist, no controller triggers)

MISSING MODULES

- Staff Password Reset
- Scheduled Auto-backup
- Queue Worker (Production)
- Report Generation
- Complete i18n

8. Panel Inventory

A. Public Website (/)

Layout: FrontendLayout.jsx (27KB)

Pages: Homepage, All Courses, Course Details, Result Verification,

License Verification, Notice Board, Video Gallery, Success Students, Verified Centres, Contact Form, Centre Application, WhatsApp Links

Features: Multi-language (en/bn/ar), feature flag toggles, referral tracking

B. Admin Panel (/admin/*)

Layout: AdminLayout.jsx (20KB) — auth:admin guard

Dashboard: Stats cards + analytics charts (monthly registrations, status breakdown, top centres)

Modules: 30+ sections covering all system data entities

C. Staff Portal (/staff/*)

Layout: StaffLayout.jsx (16KB) — auth:staff guard (Team model)

Dashboard: KPI metrics, referral link, student stats (Staff/Dashboard.jsx 28KB)

Modules: My Courses, My Sessions, My Students, Document Generation

D. Student Portal (/students/*)

Layout: StudentLayout.jsx (15KB) — auth:student guard (Student model)

Gate: portal.student middleware (toggle_student_portal feature flag)

Dashboard: Enrollment info, payment status, academic summary

Modules: Results, Documents (ID, admit, registration, marksheet), Online Exam (incomplete)

E. Centre User Portal (/dashboard, /student/*)

Layout: CenterLayout.jsx (18KB) — auth web guard (User model + center_id)

Dashboard: Basic centre information

Modules: Student CRUD (auto-scoped to own centre by CenterScope)

9. Page & Route Inventory

Total routes: ~120+

Public Routes (24)

/ | /all-course | /course-details/{id} | /verify | /result |

/license-view/{number?} | /all-notice-list | /all-notice-list/{id} |

/video-gallery | /success-student | /verified-center | /contact-us |

/center-request | /page/{type} | /health | /lang-change |

SECURITY: /live_deploy (REMOVE) | /ping | /test-transcript

Payment Routes (6)

/payment/checkout | /payment/process | /payment/callback/{gateway} |
/payment/success | /payment/failed | /payment/cancel

Admin Routes (60+) — all require auth:admin

/admin/dashboard | /admin/student (CRUD+import/export/admit/certificate) |
/admin/subject | /admin/session | /admin/exam | /admin/question |
/admin/result | /admin/center | /admin/notice | /admin/slider |
/admin/team | /admin/team-performance | /admin/financial |
/admin/grade-scales | /admin/document-templates | /admin/documents/bulk-* |
/admin/registration-review | /admin/diplomas | /admin/license |
/admin/payment-gateway | /admin/sms-gateway | /admin/contactUs |
/admin/translation | /admin/sponsor | /admin/backup | /admin/user |
/admin/sub-admin | /admin/upazila-store | /admin/whatsapp-link |
/admin/youtube-video | /admin/footer-link | /admin/footer-logo |
/admin/api-settings | /telescope

Staff Routes (15) — all require auth:staff

/staff/login | /staff/logout | /staff/dashboard |
/staff/courses | /staff/sessions | /staff/students | /staff/documents

Student Routes (15) — all require auth:student + portal.student

/students/login | /students/forgot-password | /students/reset-password |
/students/logout | /students/dashboard | /students/results |
/students/documents | /students/id-card | /students/admit-card |
/students/registration-card | /students/marksheet | /students/exam

10. Database Architecture

Total tables: ~40+ | Total migrations: 75

Authentication Tables

- admins (Admin users — Laratrust RBAC)
- users (Centre operators — web guard)
- teams (Staff agents — staff guard, also Team model)
- Laratrust: roles, permissions, role_user, permission_role, permission_user

Academic Tables

- centers (Training centres, tenant key)
- subjects (Courses — linked to team for staff scoping)
- sessions (Academic sessions — linked to team for staff scoping)
- students (All student records — CenterScope applied)
- results (Student exam marks: written + practical + viva)
- semester_results (Per-semester results for diploma students)
- grade_scales (Dynamic grading rules per course_type)
- exams (Exam definitions)
- quations (Exam questions — NOTE: typo in table name)

Financial Tables

- payments (Manual admin-recorded payments)
- transactions (Online payment records — polymorphic)
- payment_gateways (bKash/SSLCommerz credentials + config)
- sms_gateways (Reve/Twilio credentials + config)
- team_sales_targets (KPI targets per staff per month)

Content/CMS Tables

- site_configs (Single-row — all CMS + 10 feature flags)
- sliders, notices, contact_us
- translations, config_dictionaries
- youtube_videos, whatsapp_links
- footer_links, footer_partner_logos
- licenses (Credential/certificate registry)

Document System Tables

- document_templates (Canvas template definitions)
- document_fields (Per-field position, style, data mapping)

Location Data

- divisions, districts, upazilas (Bangladesh location hierarchy)

Key Relationships

```
Team → Centers → Users
                → Students → Results
                        → SemesterResults
                        → Transactions (polymorphic)

Session → Students

Subject → Students

DocumentTemplate → DocumentFields

GradeScale → used by Result::gpa()

Exam → Quations (Questions)
```

11. API Documentation

BDNSI uses Inertia.js — there is NO public REST API.

All data flows through Inertia page renders and form submissions.

The api.php file exists but contains only unused Sanctum token routes.

External APIs called by BDNSI:

- bKash: token/grant, checkout/create, checkout/execute
- SSLCommerz: gwprocess, validator/api
- Google Gemini: OCR data extraction
- Reve SMS: send SMS
- Twilio: send SMS

12. Authentication

Guard	Model	Table	Login URL	Password Reset
web	User	users	/login	Yes
admin	Admin	admins	/admin/login	Yes
student	Student	students	/students/login	Yes
staff	Team	teams	/staff/login	NOT IMPLEMENTED

All: session-based driver, bcrypt passwords

Admin login: throttled to 5 attempts/min

Student portal: guarded by toggle_student_portal flag

13. Authorization

RBAC (Laratrust) applies to Admin model only.

Roles and permissions stored in database.

Shared to frontend via auth.admin.permissions prop.

Route-Level Guards:

- auth:admin — all admin routes
- auth:staff — all staff routes
- auth:student — all student routes
- auth (web) — centre user routes
- module:toggle_X — public feature flags

GAPS:

- Most admin routes use auth:admin only, NOT permission checks
 - Sub-admin has near-equal access as super-admin in many areas
-

14. Business Workflows

Student Enrollment

Centre/Admin creates student → fill demographics + session + subject + center

→ System auto-generates roll + registration numbers

→ Status: Pending → Admin approves → Approved

→ Admin sets password → Student can login to portal

Result Entry

Admin selects student → enters marks (written + practical + viva)

→ Result::saving() validates total <= GradeScale.max_marks

→ GPA dynamically calculated from GradeScale.rules (JSON array)

→ Public result checker shows result by roll/registration/name

Document Generation (System A)

Admin selects template + student → DocumentGeneratorService renders HTML

→ html2pdf.js creates PDF in browser (client-side)

Document Generation (System B — Bulk)

Admin selects students → BulkDocumentController::bulkGenerate()

→ GenerateBulkDocumentsJob dispatched to queue

→ PdfEngineService → Node.js pdf_engine.mjs

→ ZIP created → admin downloads

[REQUIRES queue worker running on production]

Online Payment

User → /payment/checkout → selects gateway

→ Process: creates pending Transaction → redirect to gateway

→ Callback: verify with gateway API → mark success

→ PaymentSucceeded event → UpdateStudentFinancialStatus listener

→ Student.paid_amount + payment_status updated

Referral Tracking

Staff shares ?ref=STF-XXXXXX URL

→ CaptureReferralMiddleware stores ref_code in session

→ On student creation: links student.team_id to matching team

15. Frontend Architecture

Layouts (5 total)

- FrontendLayout.jsx (27KB) — Public website
- AdminLayout.jsx (20KB) — Admin panel
- CenterLayout.jsx (18KB) — Centre portal
- StaffLayout.jsx (16KB) — Staff portal
- StudentLayout.jsx (15KB) — Student portal

Large Components Needing Splitting

- Result.jsx: 52KB (largest page)
- Welcome.jsx: 43KB (homepage)
- Staff/Dashboard.jsx: 28KB
- Admin/Dashboard.jsx: 21KB

Shared Props (via HandleInertiaRequests::share())

auth.user | auth.admin | auth.staff | auth.student |

flash.success | flash.error | site_config | footer_links |

footer_logos | locale | app_url

Notable Issues

- DataTables + jQuery used alongside React — legacy dependency
 - No global state management (no Redux/Zustand) — all server-driven
 - Recharts used for analytics charts (admin + staff dashboards)
-

16. Backend Architecture

Controller Pattern

All controllers return `Inertia::render()` for page loads.

Form submissions use `redirect()->back()->with('success', ...)` or `response()->success()`.

Service Layer

- `DocumentGeneratorService` — template rendering
- `PdfEngineService` — Node.js bridge
- `FrontendDataService` — cached homepage data
- `SmsGatewayFactory` — SMS driver factory

Observer Pattern

- `StudentObserver` — fires on student CRUD events
- `ResultObserver` — fires on result save/delete

Event-Listener

- `PaymentSucceeded` → `UpdateStudentFinancialStatus`

Rate Limiting (`RouteServiceProvider`)

- api: 60/min | web: 120/min
 - results: 10/min | contact: 3/min
 - admin-login: 5/min | health: 30/min
-

17. State Management

No global state library (Redux/Zustand/etc.).

- Server state: Inertia props → React component props
- Local state: React useState/useEffect
- Form state: local state + Axios POST
- Flash messages: Laravel session → Inertia shared props
- Cache: Server-side (Laravel Cache facade)

Cache keys: homepage_sliders, homepage_courses, homepage_teams,
homepage_notices, homepage_centers, homepage_success_students,
footer_links, footer_logos, site_config_cache, unread_inquiries

18. File Management

Storage: Local filesystem (storage/app/)

Public access: Via storage:link

Image processing: intervention/image v2.7

Cast: ImageField::class auto-prepends asset URL

Upload paths:

- images/students/ — Student photos
 - images/avatar/admin/ — Admin avatars
 - center/photo/ — Centre director images
 - center/logo/ — Centre logos
 - center/nid_photo/ — NID documents
 - team/images/ — Team member photos
 - license/ — License holder photos
 - config/ — Site config images
-

19. Notifications & SMS

Email: Built-in Laravel (password reset only — no custom templates)

SMS — STATUS: 30% (not wired)

Infrastructure exists:

- SmsGateway model (DB-stored credentials)
- SmsGatewayFactory (driver factory)
- ReveSmsDriver (Reve SMS API)
- TwilioSmsDriver (Twilio)
- SendStudentSmsJob (queued)
- SendPaymentConfirmationSmsJob (queued)

GAP: No controller dispatches these jobs. SMS is never actually sent.

20. Payment System

Gateways: bKash Tokenized, SSLCommerz

Status: Integrated but not production-tested

Flow:

1. /payment/checkout → shows active gateways
2. User selects gateway
3. /payment/process → creates pending Transaction
4. Redirect to gateway
5. /payment/callback/{gateway} → verify with API
6. Update Transaction.status = 'success'
7. Fire PaymentSucceeded event
8. Student finances updated via listener

Issues:

- Credentials stored unencrypted in payment_gateways table
- SSLCommerz IPN URL not configured
- Not tested with real money

- Manual payments table and online transactions table not linked

21. Integrations

Integration	Status
bKash Tokenized API	Integrated, unverified in production
SSLCommerz	Integrated, IPN URL unconfigured
Reve SMS	Driver ready, not wired
Twilio	Driver ready, not wired
Google Gemini AI	Partial (OCR controller exists)
Hostinger VPS	Active production host
GitHub	Active version control
Laravel Telescope	Active, admin-secured

22. Security Audit

CRITICAL (Fix Immediately)

1. /live_deploy route — unauthenticated DB migration + DROP TABLE

Location: routes/web.php lines 130-140 — REMOVE IMMEDIATELY

1. Debug routes in production — /ping, /test-transcript — no auth

Location: routes/web.php lines 142-144 — REMOVE or RESTRICT

1. SSH credentials hardcoded in deploy scripts (server IP + password)

Location: direct_deploy.mjs, auto_deploy.mjs — USE .env + SSH keys

HIGH

1. Payment credentials unencrypted in DB — bKash/SSLCommerz keys at risk
2. Sub-admin authorization gaps — routes use auth:admin only, not permissions

MEDIUM

1. CORS uses abandoned fruitcake/laravel-cors package
2. Staff password reset not implemented
3. Raw DB::table() queries may bypass CenterScope
4. File upload MIME validation consistency unknown
5. CAPTCHA implemented but not applied to login forms

IMPLEMENTED (Positive)

- bcrypt passwords | CSRF protection | Rate limiting
 - Security headers (CSP, X-Frame-Options, etc.) — confirmed on live site
 - Telescope secured | Session-based auth | Inertia/React XSS protection
-

23. Performance Audit

Strengths

- Comprehensive caching for frontend data
- SiteConfig 24-hour cache
- Homepage data cached per key
- Cache auto-cleared on model save

Issues

- Welcome.jsx (43KB) + Result.jsx (52KB) — large bundles, need splitting
 - Admin/StudentController.php (32KB) — monolithic, needs service extraction
 - Dashboard queries not cached — N+1 risk on top-centres query
 - DataTables + jQuery is redundant with React
 - No Redis (likely not on Hostinger basic plan) — file-based cache
-

24. Testing Audit

PHPUnit: 31/31 tests passing

Playwright: 16 E2E spec files

Tested: Routes, feature flags, rate limiting, GPA calculation, tenant isolation, auth flows, admin navigation, student enrollment

Not Tested: Payment flow, SMS, staff password reset, permission enforcement, bulk PDF, queue jobs

25. Technical Debt

P0 — Critical (Security):

- Remove /live_deploy and debug routes
- Encrypt payment gateway credentials
- Remove hardcoded SSH credentials from deploy scripts

P1 — High:

- Wire SMS jobs to controller triggers
- Implement staff password reset
- Consistent sub-admin permission enforcement
- Setup production queue worker
- Complete student document template linking

P2 — Medium:

- Split Welcome.jsx (43KB) into section components
- Split Result.jsx (52KB) into smaller components
- Extract service from Admin/StudentController.php (32KB)
- Consolidate dual PDF systems
- Replace DataTables/jQuery with React-native solution
- Migrate CORS from abandoned package
- Complete i18n across all UI

P3 — Low:

- Add CAPTCHA to login forms
 - Remove duplicate root-level test files (test_login2.js through test_login9.js)
 - Fix table name typo: quations should be questions
 - Clean up root-level debug/scratch files
-

26. Bugs & Issues

ID	Issue	Location	Priority
BUG-01	/live_deploy route destroys DB without auth	routes/web.php:130	P0
BUG-02	Staff cannot reset password	No route exists	P1
BUG-03	SMS jobs never dispatched	No controller wiring	P1
BUG-04	Bulk PDF requires queue worker (not set up on prod)	No supervisor config	P1
BUG-05	Table typo: quations instead of questions	Migrations	P3
BUG-06	fruitcake/laravel-cors abandoned	composer.json	P2
BUG-07	fzaninotto/faker abandoned (dev dep)	composer.json	P3

27. Missing Features

Required to Finish Current Architecture

1. Staff password reset flow
2. SMS trigger wiring (student enrollment, payment confirmation)
3. Queue worker setup on production (supervisor config)
4. Student document template rendering verification
5. SSLCommerz IPN URL configuration
6. Complete online exam flow (student exam taking, submission, grading)

Recommended for Production

1. Payment credential encryption
2. Sub-admin permission enforcement
3. Remove /live_deploy route (CRITICAL)
4. GitHub Actions CI/CD pipeline
5. Automated database backup scheduling
6. Application error monitoring (Sentry or similar)

Future/Advanced Features

1. Report generation per centre/session/course

2. Student attendance tracking
 3. Fee schedules and installment plans
 4. Bulk SMS campaigns
 5. Certificate blockchain verification
-

28. Recommended Improvements

Priority order:

1. Fix 3 critical security vulnerabilities
 2. Wire SMS system
 3. Setup production queue worker
 4. Implement staff password reset
 5. Production-test payment flow
 6. Split large React files
 7. GitHub Actions CI/CD
 8. Complete Bengali translations
 9. Apply CAPTCHA to logins
 10. Consolidate PDF systems
-

29. Final Architecture Decision

PRESERVE

Multi-guard authentication | CenterScope tenant isolation |

Feature flag system | Event-Listener for payments |

Observer pattern | Inertia.js + Laravel | Laratrust RBAC |

Rate limiting configuration

REFACTOR

Admin/StudentController.php → extract to StudentService |

Welcome.jsx → split into section components |

Result.jsx → split into SearchForm + ResultDisplay |

Consolidate dual PDF systems (keep System B) |

Replace DataTables/jQuery with React table

FIX (Must Fix)

Remove /live_deploy route |

Remove /ping + /test-transcript debug routes |

Implement staff password reset |

Wire SMS jobs |

Setup production queue worker

COMPLETE

Online examination student flow |

Student document template rendering |

Financial → online transaction linkage |

Sub-admin permission enforcement |

Multi-language translations

ADD

GitHub Actions CI/CD |

Automated scheduled backups |

Supervisor queue config |

Payment credential encryption |

Redis if available

AVOID

No separate REST API/microservices |

No framework rewrite |

No additional payment gateways until existing ones tested |

No new features until security patched

30. Development Roadmap

Sprint 1 — CRITICAL (Week 1)

- Remove /live_deploy and debug routes
- Encrypt payment credentials in DB
- Move SSH credentials to environment variables
- Setup production queue worker

Sprint 2 — HIGH (Weeks 2-3)

- Wire SMS triggers (student enrollment + payment)
- Implement staff password reset
- Complete student document template linking
- Configure SSLCommerz IPN URL

Sprint 3 — MEDIUM (Weeks 4-5)

- Complete online examination student flow
- Sub-admin permission audit + enforcement
- GitHub Actions CI/CD pipeline
- Automated backup scheduling

Sprint 4 — POLISH (Weeks 6-7)

- Split large React components
 - Complete Bengali translations
 - Apply CAPTCHA to logins
 - Payment sandbox testing
-

31. AI Agent Rules

MANDATORY PRE-CHANGE CHECKLIST

1. Read this document
2. Read the relevant section in PROJECT_DOCUMENTATION/
3. View the source file with view_file before editing
4. Check related models, routes, controllers, frontend pages
5. Check CenterScope implications (tenant isolation)
6. Check feature flag implications (module toggles)
7. Make the smallest safe change
8. Run: C:\xampp\php\php.exe artisan test (all 31 must pass)
9. Run: npm run build
10. Commit + push + deploy

CRITICAL CODE RULES

GUARD USAGE:

NEVER: `auth()->user()` (without specifying guard outside web context) ALWAYS specify: `auth('admin')->user()` — admin context `auth('staff')->user()` — staff context `auth('student')->user()` — student context `auth()->user()` — ONLY for web (centre user) context

TENANT ISOLATION:

NEVER bypass CenterScope in centre-user context NEVER use `DB::table('students')` without manual `center_id` filter Student and Result have CenterScope global scope applied

FEATURE FLAGS:

New toggleable features need: 1. Column in `site_configs` via migration 2. Added to `SiteConfig::$fillable` 3. Added to `HandleInertiaRequests::share()` `site_config` array 4. Toggle UI in admin settings 5. - `>middleware('module:toggle_column')` on route

ENUMS — Always use enum classes:

`StudentStatus::Pending / ::Approved / ::Hide` `Gender::Male / ::Female` `CourseType::Regular / ::Short_Course / ::Diploma` (and all other enums in `app/Enums/`)

IMAGES:

Store via `Image::store($field, $folder)` — returns relative path NEVER store full URLs in database

CACHE:

After modifying model data, cache auto-clears via ClearsFrontendCache For manual cache clear: php artisan optimize:clear

PAYMENT:

NEVER mark transaction 'success' without gateway API verification ALWAYS wrap payment in DB::beginTransaction() / commit() ALWAYS fire PaymentSucceeded event after successful payment

WHAT NEVER TO DO:

Never hardcode center_id values Never store credentials in code — use .env Never skip CSRF on forms Never commit .env files Never run migrate:fresh on production without backup Never create duplicate models (check app/Models/ first) Never create duplicate routes (check routes/*.php first) Never leave /live_deploy route in any code

32. Final Audit Summary

Project Statistics

Metric	Count
Total modules	35+
Total panels	5 (Public, Admin, Staff, Student, Centre)
Total React pages	70+
Total routes	120+
Total database models	35
Total database tables	40+
Total migrations	75
Auth guards	4
Feature flag toggles	10
PHPUnit tests	31 (all passing)
Playwright E2E specs	16
Major business workflows	6

Summary

- Completed: Public website, admin CRUD, enrollment, results, document builder, payment integration, Telescope, tests
- Partial: Online exam, SMS, financial, sub-admin RBAC, staff portal
- Missing: Staff password reset, queue worker, CI/CD, production-tested payments
- Security: 3 critical, 2 high, 5 medium issues identified
- Technical debt: 3 P0, 5 P1, 7 P2, 5 P3 items

TOP 10 NEXT ACTIONS

#	Priority	Task	Reason
1	CRITICAL	Remove /live_deploy route	Anyone can destroy DB
2	CRITICAL	Encrypt payment credentials in DB	bKash/SSLCommerz keys exposed
3	CRITICAL	Remove SSH credentials from deploy scripts	Server root access exposed
4	HIGH	Setup production queue worker (supervisor)	Bulk PDF + SMS silently fail
5	HIGH	Wire SMS triggers to student events	Core feature not working
6	HIGH	Implement staff password reset	Locked-out staff need admin help
7	MEDIUM	Test payment flow in sandbox	Revenue feature unvalidated
8	MEDIUM	Complete student document rendering	Student portal incomplete
9	MEDIUM	Build GitHub Actions CI/CD	Manual deploys are fragile
10	MEDIUM	Complete online examination flow	Feature in progress, unusable

Document created: 2026-08-24

All findings verified from direct source code inspection.

No features were assumed or invented — only documented facts.

See PROJECT_DOCUMENTATION/ folder for detailed coverage of each area.